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Designing systems in a fine-grained and inclusive way, by integrating Service Design within Futures Studies and Foresight

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This paper evaluates the human-centric, systemic and near future-driven components of Service Design to understand how they can work within a Future Studies and Foresight (FSF) framework. Expanding on empirical evidence gathered from a case study with renters and shareowners in Danish cooperative housing associations (Andelsboligforeningen), the paper looks at how the application of Service Design – within Future Studies and Foresight (FSF, hereafter) – can facilitate a fine-grained and inclusive rebuilding of our socio-technical systems in a way that allows end-users to understand and engage with them more productively. The paper provides a theoretical reflection on the gestalt emerging from bridging Service Design and FSF and how this can inspire and establish strategic viewpoints on how to co-design complex systems embedded in different socio-technical dimensions.

***Keywords**: Service Design, Futures Studies and Foresight, Systemic Value Creation.*

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1. Introduction

This paper starts with the premise that worldwide cultural and societal changes made us realise how wicked problems require a more systemic problem-solving perspective to create preferable and desirable futures for people, institutions, and organisations. Also, COVID-19 has shown the world the need to rethink planning processes by inclusively empowering human societies (Jon, 2020).

Cultures are already being redefined: thought-provoking interactions are forcing technocratic systems to become critical of themselves, such as the case of hashtag activism on social media (Yang, 2016). Jones (2014, p. 120) argued, "Social participation reinforces the self-organisation of co-created content and purposeful interaction within the boundaries and norms of the social system". When discussing the role of design in the context of socio-technical systems, this research resonates with the perspective proposed by Escobar (2018), reflecting on the ontological shift of design from its functionalist and industrial tradition towards a more relational dimension of life. This relational perspective, firmly attached to systems change, paves the way for strategic approaches to redesign our cultures while addressing positive change.

It is in this context that this research advances the role of Service Design, as a practice rooted in design thinking (Brown, 2008) and working in a human-centred way (Stickdorn et al., 2018), to dialogue with the immaterial components of systems and incorporate both present and future realities in the creative process (Penin, 2018; Morelli et al., 2021). Various academics (Brown & Katz, 2009; Manzini, 2015) suggested to us how, in a world where everybody designs, Service Designers can bring the right tools to those people who have never thought of themselves as designers, allowing them to make sense of their capabilities as means to influence their own systems. By applying co-creative approaches to design with a clear human-centred perspective (Manzini, 2015), Service Designers can become agents for redesigning cultures and decision-making processes at individual and more complex institutional levels (Hill, 2012).

The possibility of Service Design to work with changing systems – benefitting from an unpredictable world as a fuel for creative thinking and doing – makes us reflect upon other practices that have already embarked on such an effort; this is the case of Future Studies and Foresight (FSF). As defined by Conway (2021), FSF is meant to create better organisational structures and processes to tackle rising uncertainties. While some design and FSF researchers and theorists tried to juxtapose FSF and design thinking as an integrated methodology (Hines & Zindato, 2016; Gordon et al., 2019; Bühring & Liedtka, 2018) and others reflected upon the critical co-creative potential of foresight methods (Wilkinson et al., 2014; Bourgeois et al., 2017; Miller, 2015), many FSF practitioners (Waverley Consultants, 2017; Dyrman et al, 2018; Smith & Ashby, 2020) noted the lack of human-centred perspectives within standard FSF practices.

This paper addresses this gap by exploring how Service Design and FSF might support one another, thus providing the possibility to (1) map the functioning of systems at a more granular level and (2) make clear how individuals and their actions can affect these systems.

The paper is structured as follows. In Section 2, we look more deeply into FSF and its limitations; an initial understanding of how Service Design works with futures will be followed by the linkage between FSF and Service Design. While in Section 3, we introduce the case study and describe the research methods, in Section 4, we draw from the empirical material collected within the context of the Danish *Andelsboligforeningen*. Key considerations emerging from both empirical and theoretical viewpoints will be discussed in Section 5, representing also the conclusion of this paper.

2. Literature Review

2.1 Future Studies and Foresight and Limitations

FSF is mainly concerned with studying change: systems, organisations and complex interactions are all subjects of change and, therefore, become subjects for future studies (Bishop & Hines, 2012). Wendy L. Schultz (2015), in the article 'A Brief History of Futures', retraces the different phases of FSF throughout different moments in time. Schultz (2015) argues how after WW2, the blossom of interest in future thinking was supported by the increasing attention toward innovative practices to plan for complex systems and

shattered infrastructures (Schultz, 2015; Conway, 2021). FSF's contribution became very helpful over the years in planning for a disorganised world; companies and organisations found support within FSF methods and tools to plan and innovate for better technological, economic and even political realities (Conway, 2021).

In the meantime, design thinking also gained momentum among FSF practitioners and researchers; participatory field research, user journey and personas are just some of the design tools borrowed by FSF to advance desirable visions of the future (Waverely Consultants, 2017; Dyrman et al., 2018; Smith & Ashby, 2020). Despite the influence of participatory approaches within FSF, the mainstream idea remained anchored to a limited inclusion of high-level stakeholders in design processes (Nikolova, 2013), as, for example, with the case of the Delphi methods (Waverley Consultants, 2017). This consideration highlights how, despite the engagement of experts, other actors affected by the local context remain excluded from the possibility to engage. Practitioners such as Dyrman et al. (2018) or Smith and Ashby (2020), while being very good at systematically detecting signs of any potential development through methods like Horizon Scanning, do not emphasise the importance of user involvement to detect patterns of coming disruption. As proposed by Nikolova (2013, p. 5), to strengthen the participatory aspect of FSF, there is the need to "allow citizens, be they stakeholders, non-specialists, and concerned individuals, to impact the process."

Another significant limitation of FSF concerns the challenges that emerge while trying to make sense of future visions. FSF works with complexity, yet the definition of a clear and granular vision of how the future will unfold may result in many attempts from futurists to point out future opportunities and threats (Hines & Zindato, 2016; Conway, 2021). Hines and Zindato (2016) suggest how FSF should also take a step forward and try to help organisations do something about future visions, facilitating the comprehension of systems to foster change. In many cases, such as the way futurists employ Scenario planning, the dynamics of intertwined human and planetary systems are indeed a primary concern for FSF (Conway, 2021), which should expand its core methodology to advance the implementation of change (Bishop & Hines, 2012).

Besides not being the primary focus of this research paper, systems thinking is a precondition to better understand the gestalt that may result when combining Service Design and FSF in a joint intervention. Considering the nature of those fields, systems thinking is necessary to situate this work and convey the limitations of FSF and the possible contribution of Service Design. Systems analysis is essential to predicting complex dynamics in social systems. Geoffrey Vickers, a pioneering system scientist, suggests how systems analysis is an operational part of systems thinking, which does not necessarily reveal what problems could or should be addressed by human action to change complex situations (Vickers, 1983). As cited in Buchanan (2019), Vickers suggests that to better understand problem-solving for systems, it is necessary to rely on human experiences and local circumstances – contrary to a more technocratic problem-solving perspective from FSF. As also proposed by Pereno and Barbero (2020), systems transition – mainly when referring to territorial systems – occurs when a design approach is built upon the understanding of systems dynamics, with a multi-level approach to design as well as a solid co-creative approach that involves different types of stakeholders directly concerned with the system itself.

Service Design employed to inclusively and granularly operationalise systems change

Polaine et al. (2013) argue that designing for services also means designing the relationships between people inside or outside a specific system. As such, services are conceived with people at the centre. This paradigm, thoroughly discussed by design theorists such as Ezio Manzini (2015), advances the proposal for a more plural, participative and diffused role of design in society. The rising complexities surrounding design interventions grew together with an ontological approach to design (see Escobar, 2018), which points out that nurturing transitions and change requires a reorientation from the industrial and technocratic tradition whence it originated. The shift away from its functionalist perspective positioned design in a transition from the production of goods towards the production of knowledge-intensive systemic solutions (Morelli, 2002; Butera, 2005), where participatory traditions such as the Scandinavian Participatory Design approach from the 1970s became very valuable. Inspired by these influences, Simonsen and Robertson (2013), as cited in Ehn et al. (2014, p.5), claimed that "if we are to design the futures we wish to live, then those whose futures are affected must actively participate in the design

process". The engagement of users in the process of design would make such a process more inclusive; the need to collaboratively envision desirable futures became an essential requirement for all those interventions related to social design and democratic design that find fertile ground in urban planning design processes (as in the case study of this paper).

When analysing change, FSF and Service Design provide different approaches and goals. As underlined by Kulsbjerg Løgager et al. (2021, p. 3), while both Service Design and FSF are future-oriented, "by working towards and around something which does not already exist", FSF analyses long-term views, and Service Design focuses on the creation and deployment of services within a near-future perspective. It is possible to find this orientation in how designers and futurists apply Scenario planning exercises. Maschi (2002) suggests how the designer's point of view on Scenarios follows a scope, concept and solution-based approach, envisioning tools to implement and visualising specific solutions within systems. Futurists, instead, focus on a more mission and context-based approach to draw stakeholders' attention towards a common goal or explore alternative contexts through narratives and storytelling (Maschi, 2002). Besides being theorised multiple times and in different contexts, this research assumes that a bridging mechanism between FSF and Service Design can be found in how both fields work with uncertainty and, therefore, with methods like Scenario. The lens by which a concept-based approach can work with a more explorative and context-driven one finds fertile ground with the co-creative and systemic nature by which Scenario has been applied (Ogilvy, 2002; Wilkinson et al., 2014; Miller, 2015; Bourgeois et al., 2017). By linking the way systems can be analysed (concept of System Analysis by Vickers, 1983), together with a more precise definition of systemic structures (Morelli et al., 2020) or interdependencies (Buchanan, 2019), it might be possible to support value creation more inclusively and granularly.

The effort that is required to explore the added value that might emerge from the integration of Service Design within FSF can be encountered in the formulation of the following research question:

How might Service Design support Future Studies and Foresight to design systems in a fine-grained and inclusive way?

3. Methodology

The urban context harbours a plethora of complex socio-technical systems in which design has been showing interest (Pereno & Barbero, 2020). These geographical realities have a complex nature determined by different social, cultural and technological dynamics – and indeed, this research inquiry can be well settled within the urban context of cooperative housing. On another note, preliminary desktop research on cooperative housing pointed to the challenges that unorganised communities face, mainly resulting in a lack of inclusion and citizen engagement (Williams, 2005; Vestergaard, 2007; Lang, 2015; Larsen, 2019).

To investigate cooperative housing, we focused on a case study (Yin, 2018), unfolding across a period of 5 months and anchored to the emerging challenges of democratic engagement and participation within *Andelsboligforeningen* communities in Copenhagen, Denmark. The case helped us understand how cooperative housing associations are organised and how people could become more included in the decision-making processes by engaging more deliberately with the several layers of complexity shaping their cooperative living context. The role of a case study in such a context supports an evidence-based approach to design; however, it is worth mentioning how – being a single case study– this case manifests multiple limitations regarding timing, resources, and scalability. A Research Through Designing approach (RTD – Lenzholzer & Brown, 2016) has been employed within the case. Activities and experiments play a key role in conducting research through design; on the one hand, they allow to address specific problems of concern; on the other, they allow to intuitively reflect upon specific aspects of theories and approaches within the design context.

The case can be broken down into two main steps: 1) data collection, where desktop research has been incorporated with individual/group semi-structured interviews, involving both share owners as part of a cooperative association (n=4) as well as shareowners also members of different cohousing's boards of administration (n=2); 2) the facilitation of a co-creative Scenario experiment together with different

individuals from different housing cooperative associations in Copenhagen (n=6); some of them have been previously engaged during the data collection phase.

The hybrid framework used in this study: FSF informed by Service Design

The work conducted by Bishop and Hines (2012), as well as Gordon et al. (2019), inspired the background knowledge for the development of a hybrid framework merging both Service Design and FSF throughout a futuring process. The application within the case study culminates in developing different open-ended and co-created scenarios for possible housing cooperatives configurations in Copenhagen, Denmark. The framework is iterated in 5 different phases (Table 1); each of them highlights both FSF and Service Design methods and tools. Some of the FSF tools and methods – like the Scope Wheel and the Scan Cards, or Insights Generation - derive from the work of practitioners (Dyrman et al., 2018; Smith & Ashby, 2020). In most phases, a Service Design approach is integrated with these methods and tools.

	Phase	**Output**	**Tools/Methods**
1	Assessment	Assessment of the future exercise in terms of resources and background knowledge;	Research Plan; Scope Wheel;
2	Perceive & Empathise	Gathered user research as well as macro-externalities (trends, signals of change) with a broader perspective concerning the context;	Desk Research; In-depth interviews; Scan cards; Stakeholder map;
3	Define & Prospect	Clustered research into insights; Identification of future users and System Analysis;	Insights Generation; System Analysis; Personas;
4	Develop	Development of future visions of housing cooperatives that considers users' perspectives and the components of the cooperative as a system;	Co-creative Scenarios;
5	Deliver	Systemic representation of different scenarios resembling future visions and present needs and users' aspirations;	Scenario Evaluation;

Table 1 FSF informed by Service Design - methodology

Figure 1 shows how within the framework that we followed, methods and tools were drawn by either Service Design or FSF. In this way, it is possible to see at which stage different approaches have been utilised and how the framework bridges both fields towards a collaborative experiment advancing the creation of different outputs.

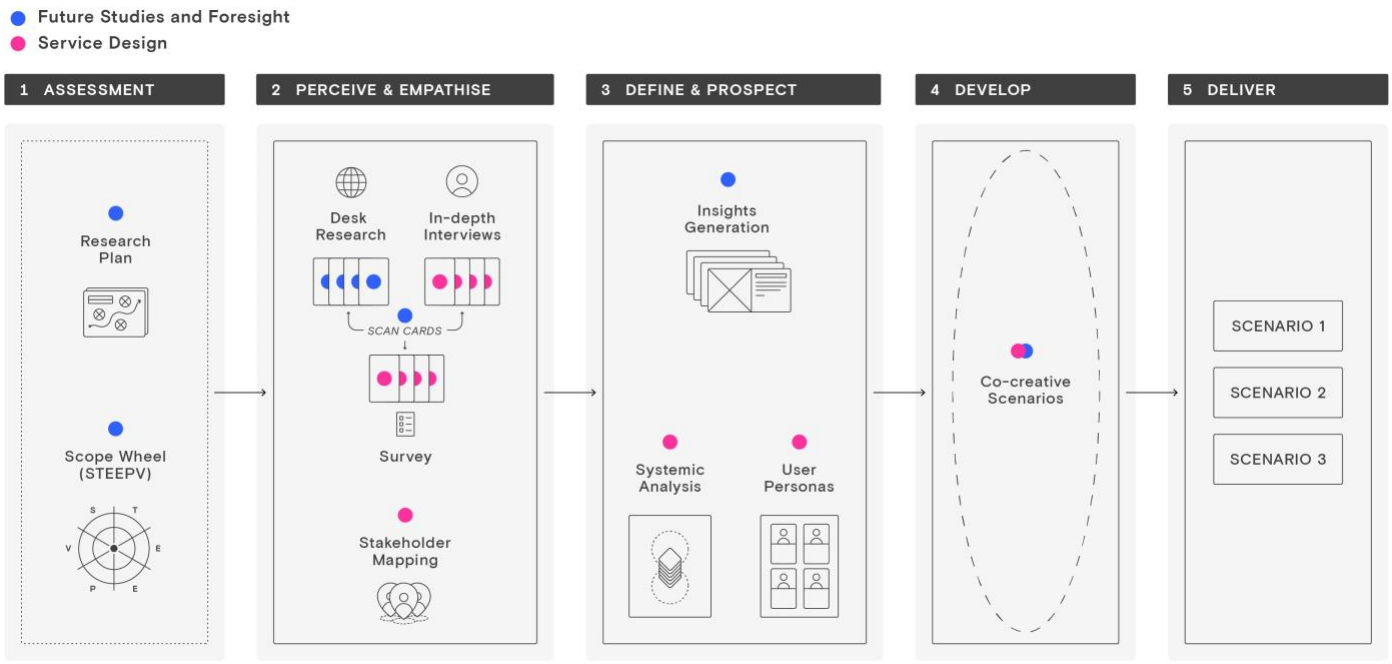


Figure 1 FSF informed by Service Design - a framework with methods and tools from FSF and Service Design.

4. Findings

In this section, we highlight where and how the role of Service Design empowered an FSF-driven process. This can be encountered in the data collection and the preparation for the Co-creative Scenario workshop. Particularly, our analysis focuses on three core aspects of our framework: (1) the inclusion of multiple viewpoints proposed during the data collection, (2) the process of sense-making where visual facilitation unravels the complex functioning of the Danish housing cooperative system, and (3) how multiple perspectives are integrated and visualised within a Co-creative Scenarios experiments, that functions as a bridging mechanism between the two arenas of Service Design and FSF.

Design-driven collaborative visioning

This framework enhanced inclusivity by looking at how it was possible to integrate – and engage with – several viewpoints into different steps of the design process. A multimethod approach (Friedman & Muratovski, 2022) allowed us to explore both secondary and primary sources for data collection; from one side, we perceived emerging trends and signals shaping the housing cooperative context; from the other side, we empathised with the users' viewpoints within the changing context of urban housing cooperatives associations in Copenhagen. A Horizon Scanning approach (Dyrman et al., 2018; Smith & Ashby, 2020) allowed us to review published sources on the topic and case studies. This approach allowed us to situate the context within a broader setting; for instance, different macro-trends highlighted how economic and societal factors encouraged people within different age groups to move into a housing cooperative in the last 10 years. This approach also allowed us to consider a broader and systemic picture; for example, we shortly encountered different trends pointing toward the change of organisational models of some cooperatives that pursued a flatter hierarchy to mitigate issues emerging from vertical power structures. Despite being able to evaluate opportunities for change within the context of housing cooperatives, we promptly realised that Horizon Scanning was not enough to cover the qualitative – and human-centred aspects of the local context where associations are established. Questions like 'What is the process for accessing a cooperative? How are they administered? How do members communicate?', as well as possible gaps, issues or opportunities emerging from current experiences, required a more relational and human-centred approach.

At this point, different semi-structured interviews (30-60 minutes) were conducted with different target groups. Following a maximum-input sampling (Stickdorn et al., 2018), we found new leads covering a wider range of interest as the fieldwork unfolded. Community administrators, international and local owners, young couples (representing one of the primary target groups), and renters were interviewed. During some of the interviews, new residents remarked on the challenges in accessing and onboarding cooperative housing associations; they claimed that "*Housing cooperatives are not equal, the market is moving fast and entering a community is much more difficult without a proper network*" or even "*it is difficult to engage with the community planning since the board of administration does not involve all residents and decisions are taken within close circles*". We also analysed housing cooperatives' onboarding process through User Journeys (Stickdorn et al., 2018), which shed light on how residents access such communities and made us reflect on how a process like this could be better designed. Another interview with one community administrator showcased the hidden structures of a housing cooperative, revealing how they are organised and how dwellers communicate with each other. This interview highlighted the residents' common interests and stakes: "[...] *one of the most important things that the dwellers care about is the market evaluation of their shares; a higher market evaluation could potentially make them richer from day to night [...] Administrators have a huge responsibility in deciding when and with whom to make the value assessment*".

While relying on a survey distributed within the network of individuals interviewed during the data collection, we systematically collected and visualised emerging trends and local viewpoints through Scan Cards (Dyrman et al., 2018; Smith & Ashby, 2020). Scan Cards (Figure 2) represented for us a way to further utilise the data throughout the other phases of the design process and facilitated the creation of more structured Insights.

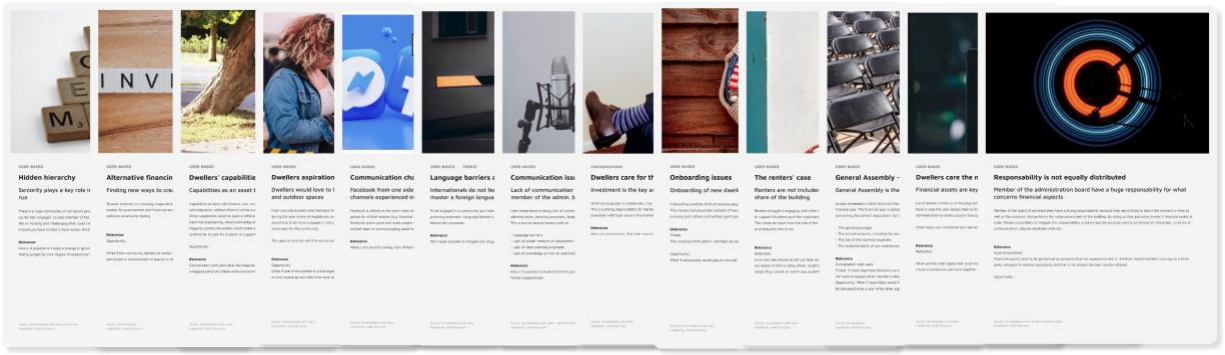


Figure 2. Example of a Scan Card cluster, here representing the primary data sources.

Engaging with local administrators of housing cooperatives in Copenhagen helped us to define how *Andelsboligforeningen* tend to be structured, considering different actors and stakeholders involved in the community constellation. Consequently, a stakeholder map of an urban cooperative housing association was developed (Figure 3) to be further utilised during the co-creative Scenario.

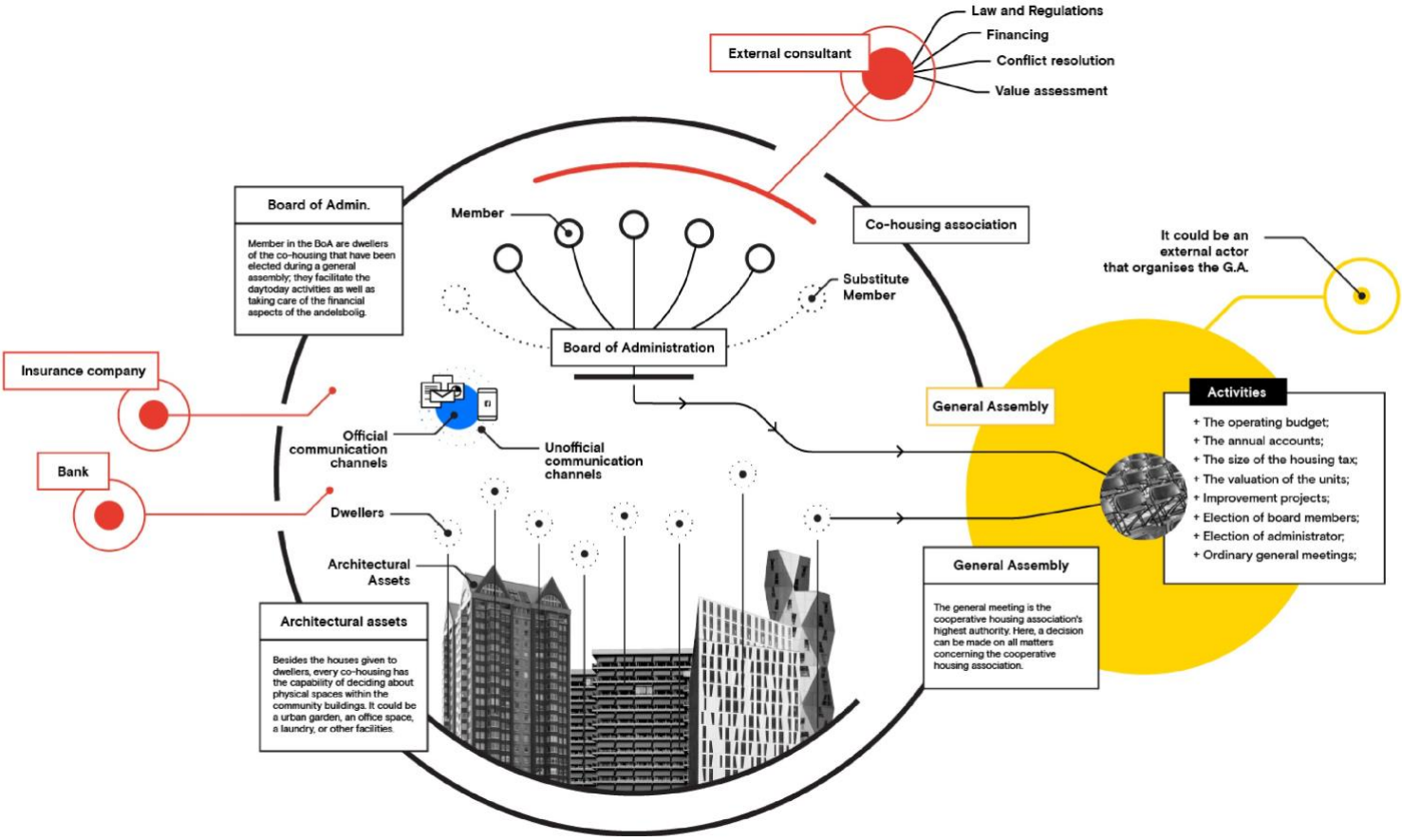
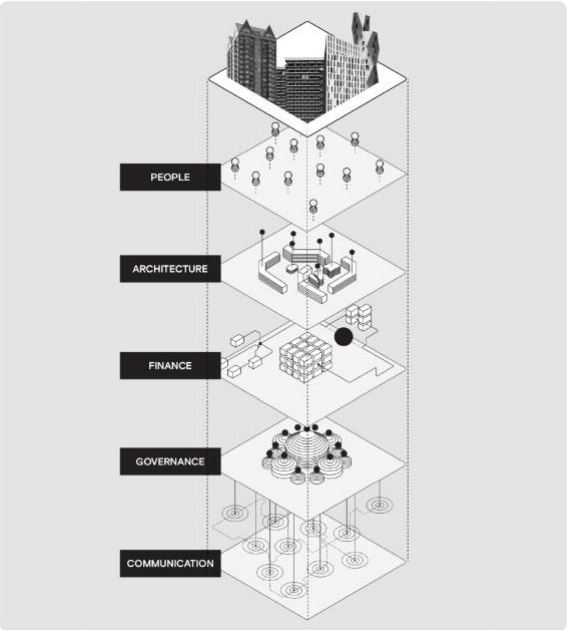


Figure 3. Stakeholder map of an urban Andelsboligforening – the map approximates the stakeholders' constellation.

Visual facilitation geared toward understanding complexity

The sensemaking phase we conducted allowed us to streamline the gathered research and reflect on how housing cooperatives are more than just housing structures populated by residents. We took inspiration from the Dark Matter Labs' (2022) article on visualising and communicating complexity, which coincidentally drew from Donella Meadows' theory for the change in systems, as presented in her famous article on leverage points (Meadow, 1999); as well as from the visualisation approach encountered in Bratton's concept of The Stack (Bratton, 2015). The analysis of such theories made us realise how systems are represented by different layers, communicating with one another and in which leverage points (or points of power) can be identified to spot systemic strengths and weaknesses. We could position many of the Insights generated within the co-housing layered system (a structure of *community assets*). Such a layered system approach follows what Morelli et al. (2020) define as structure within socio-technical systems – or even what Buchanan (2019) defines as interdependencies within systems. The visualisation with 'stacked' layers (Figure 4) helped us understand how a housing cooperative association works, and it became a visualisation tool to make such a system more granular, where value creation is situated within different assets of a specific structure. The *architecture*, *the finance models*, the *governance* and the *communication* channels have been perceived as the assets of a housing cooperative community, also influenced by the different *capabilities* of potential users within the context. Visualising the research findings – e.g., Scan Cards from the data collections and the *community assets*, allowed us to envision the complex nature of the context systematically and set the stage for communicating it to the users. To take this a step further, we decided to create future User Personas (Stickdorn et al., 2018), which have been developed by following different archetypes of target groups (elderlies, renters, Danish owners, international owners), as well as the needs and aspirations that have been gathered during the data collection phase.



PEOPLE: User's capabilities	What are those capabilities that can improve collective wellbeing? And who triggers them?
ARCHITECTURE: Physical facilities	What are the physical facilities where people interact?
FINANCE: Financial Models	What is the financial interest of dwellers? How does the community sustain itself in time?
GOVERNANCE: Organisational structures	How does the organisation take decisions?
COMMUNICATION: Channels	What are the official and unofficial channels of commuinication?

Figure 4. Community Assets and description.

After the Define & Prospect phase (see Table 1), the emerging outputs set the preconditions to situate the main experiment for the Develop phase. Such outputs increased the knowledge around how these associations work and encouraged empathy-building between the researchers and the users (later involved during the co-creative Scenarios). Conceived as boundary objects (Bjögvinsson et al., 2012; Stickdorn et al., 2018), we used them to facilitate a relational conversation between people with different backgrounds and experiences while supporting a participatory Scenario experiment.

The Develop phase of the framework was built upon two main pillars: the Scenario preparation and the Scenario activity. The preparation included the stage setting for the workshop experiment. Despite the users being directly involved in their housing cooperative community, they still needed help understanding the structure and their role within it. A Service Design approach in this regard advanced the comprehension of their role in the context. It made them understand how their environment relied on systemic pillars where their contribution could have been relevant. As shown in Figure 5, different artefacts (or Boundary Objects) followed different requirements to ease the comprehension of the *Andelsboligforening*'s ecosystem to potential users.

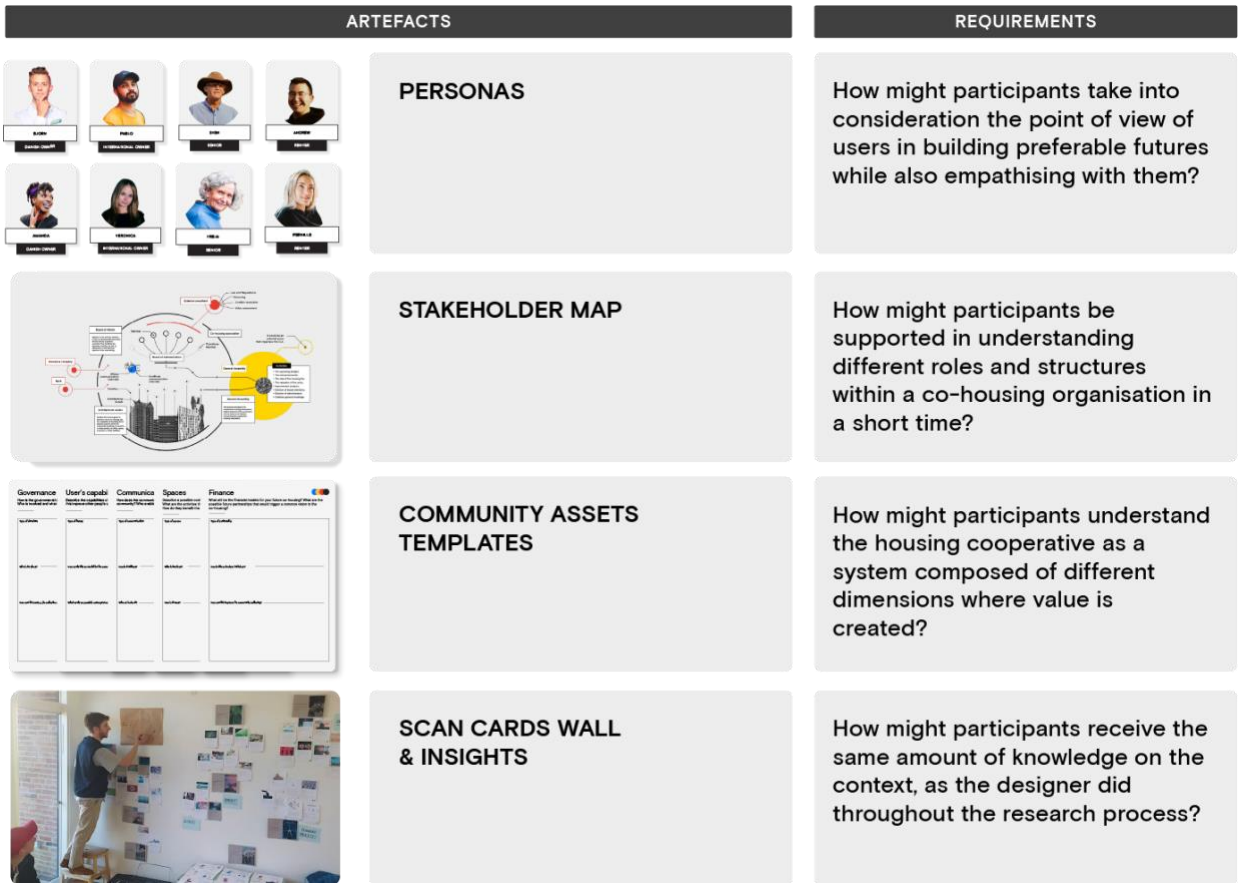


Figure 5. Co-creative Scenario workshop preparation; different artefacts for different purposes

During the Scenario experiment, participants – already engaged during the data collection that happened weeks before the experiment – were undoubtedly familiar with the context. Still, the workshop introduction was a good moment to refresh their mind and showcase the Service Design's visualisation capabilities. As shown by Dyrman et al. (2018) or Smith and Ashby (2020), a standard FSF process would have expected the only integration of Scan Cards as the process visualisation tool. However, a Service Design approach within FSF advanced the adoption of different Boundary Objects (Stickdorn et al., 2018), which have been utilised to translate the complexity of the context into a more coherent and granular subject for users belonging to different housing cooperatives.

Peculiar is the example of the Stakeholder Map, which was drawn during the workshop activities, to explain how a housing cooperative association is structured and works. This allowed the facilitator to clarify different roles within an urban housing cooperative community. Eventually, some of the participants pointed out how "*the on-the-spot explanation of the stakeholder map and the community assets enhanced my understanding of how the housing cooperative is built and what could be my role within such a community*" and even in consideration of the utilisation of User Personas one of the participants claimed that "*having the point of view of the personas allowed me to empathise with them and project my fears and ambitions within a co-housing community*".

Co-creative Scenarios as an integrating mechanism

During the Scenario experiment, we deliberately tried to integrate insights emerging from the previous phases ("Design-driven collaborative visioning" and "Visual facilitation geared toward understanding complexity"). After the workshop preparation and introduction, the Scenario activity started (see Figure 6); users (n=6) were divided into 3 groups and were given the chance to work with different templates (Community Assets templates) for the creation of scenarios. Each group explored different Community Assets and matched them with different Scan Cards while considering the User Personas capabilities.

This allowed them to interpret each asset in a human-centred way. For example, the group that envisioned elders as the primary users of the cooperative (Scenario 2) also suggested how the communication channels within a community should have been kept simple, limiting the dependency on digital technologies. Again, in Scenario 3, the group envisioned how important it was to invest in physical facilities as onboarding stations for newcomers unfamiliar with how the housing cooperative works.



Figure 6. Participants of the workshop while looking at the different Scan Cards, the relative Insights, and the cards with different User Personas.

The resulting Scenarios – refined in a second step during the Deliver phase, highlighted both design-driven and future-driven characteristics. They showcased the exploration of alternative contexts through storytelling (Context-based Scenario) and the identification of possible service concepts (Scope, Concept and Solution-based Scenario, as supported by Maschi, 2002). For example, in the output of Scenario 1, there was the need to implement a service to improve the self-organisation of the community and enhance the way citizens voted for strategic proposals (e.g., the building of physical facilities). Simultaneously, in Scenario 3, the exceeding food produced by the community's greenhouse could have been sold through a local food market hosted outside the community buildings.

When asked about feedback and reflections on how the workshop was conducted, some participants reflected on how Scenario encouraged them to become more active in their context; one of the participants claimed, "*Being part of this process allowed me to understand more the context where I live. It also motivated me to become more engaged with the community and the board of administration that runs my cooperative*". Some others even suggested how being involved in the process from the beginning made them decide to become members of the board of administration of their cooperative.

5. Discussion

Acknowledging the several limitations (time, resources and scale) that a single case study presupposes, the inductive approach emerging from the data collection and the experiment activities gave us a chance to derive hypotheses on the role of Service Design within an FSF framework. Our take is that Service Design can support Future Studies and Foresight to design systems in a fine-grained and inclusive way through three mechanisms: (1) acquiring inclusive perspectives by integrating both external and localised viewpoints, (2) planning and pre-assessing future states through visual facilitation and (3) co-creating Scenarios to address systemic value creation.

Acquiring inclusive perspectives by integrating both external and localised viewpoints

This research looked at inclusivity from both vantage points anchored within FSF and Service Design methods and tools. FSF's Horizon Scanning method for data collection presents an inclusive approach to identifying broader trends and signals of change; however, there is a growing recognition (Bishop & Hines, 2012; Hines & Zindato, 2016; Waverley Consultants, 2017; Dyrman et al., 2018; Smith & Ashby, 2020) that FSF's research angle should also encompass more localised issues. Correspondingly, the human-centred approach of Service Design stresses the importance of including users and their perspectives within the design process (Simonsen & Robertson, 2012; Jones, 2014; Manzini, 2015). However, while identifying user needs, issues, and aspirations is paramount, common Service Design approaches run the risk of being too much anchored to the local specificities of the context at hand and, as such, of missing the opportunity to zoom out, foster a perspective that values external viewpoints and reach a broader understanding of how a system might evolve over time. FSF provides this broader vantage point and integrating the two disciplines makes the whole design process both future-proof and more inclusive.

Planning and pre-assessing future states through visual facilitation

This study showed how FSF could employ different capabilities from Service Design, drawing from methods used within systemic contexts to enhance the visualisation and comprehension of complex systems. The capability of visualising and mapping complexity (Morelli et al., 2021) – possibly through boundary objects (Stickdorn et al., 2018) - is critical in representing systems to a wide variety of stakeholders, not necessarily speaking the same language and/or sharing the same needs and wants. By combining FSF-derived sense-making tools (like Scan Cards) with Service Design visualisation tools (such as Community Assets, Stakeholder Maps, and User Personas), designers can effectively map key components, roles and relationships within a system, identify leverage points for interventions and adopt a forward-looking approach that – through reasoning based on a multiplicity of what-if – supports communities in imagining and pre-assessing possible future states of complex systems (Meadows, 1999; Bratton, 2015; Buchanan, 2019; Morelli et al., 2020; Dark Matter Labs, 2022). Providing a fine-grained approach in the representation of systems enables users to grasp various levels of complexity and nurtures – also, in this case – a more inclusive and democratic understanding of challenges and opportunities.

Co-creating Scenarios to address systemic value creation

These fine-grained and inclusive representations of systems were integrated by Scenarios, which worked at a more precise level to imagine how very practically key resources and assets could be mobilised to reach future states. Scenarios have been utilised by both futurists and designers, albeit with distinct methods of conception and execution (Maschi, 2002; Hines & Zindato, 2016). During our study, it became evident that the designers' use of Scenario experiments (solution-oriented and concept-based) can complement the approach generally taken by futurists (broader, less context-specific). Within our case, the combination of the two approaches allowed the stakeholders to imagine and enable value-creation mechanisms within the system. These processes adopted a co-creative and collaborative stance (Ogilvy, 2002; Bishop & Hines, 2012), thus engaging a wide variety of participants, who felt more distinctively their agency on possible future states of the system. Merging a solution-oriented approach with a broader one can add granularity to future realities, making them more tangible and operationalisable (Hines & Zindato, 2016; Conway, 2021). This Scenario-based process of user involvement to address issues and opportunities within local systems (Vickers, 1983) follows the research advanced by Miller (2015) suggesting how Scenarios can and should empower the agency of communities in their local contexts.

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